

lec 10

Recap

Linearity of expectation $E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i]$

examples: Random, hypergeometric,
hashing, coin collection problem

Not always true that

$$E\left[\sum_{i=1}^n X_i\right] = \sum_{i=1}^n E[X_i]$$

① True if X_i are non-negative

$$\sum_{i=1}^n E[X_i] < \infty$$

② Conditioning

$$P_{X|A}(a) = \frac{P(\{X=a\} \cap A)}{P(A)}$$

$$P_{XY}(x,y) = P_{X|Y}(x|y) = \frac{P_{XY}(x,y)}{P_Y(y)}$$

$$\sum_{x \in \mathcal{X}} P_{X|A}(a) = 1$$

$$\sum_{x \in \mathcal{X}} P_{X|Y}(x|y) = 1$$

③ Multiplication rule

$$P_{XY}(x,y) = P_X(x) P_{Y|X}(y|x) = P_Y(y) P_{X|Y}(x|y)$$

$$P_{X_1, \dots, X_n}(x_1, \dots, x_n) = P_{X_1}(x_1) P_{X_2|X_1}(x_2|x_1) \dots P_{X_n|X_1, \dots, X_{n-1}}(x_n|x_1, \dots, x_{n-1})$$

Bayes rule

$R(\cdot)$

Given PMF of R.V X_1 and conditional PMF $P_{Y|X}(\cdot|\cdot)$. Find $P_{X|Y}(x|y)$

$$P_{X|Y}(x|y) = \frac{P_{X,Y}(x,y)}{P_Y(y)}$$

$$P_Y(y) = \sum_{x \in \mathcal{X}_0} P_{X,Y}(x,y)$$

$$= \sum_{x \in \mathcal{X}_0} P_X(x) P_{Y|X}(y|x)$$

$P_X(x)$

$$P_{X,Y}(x,y) = \frac{P_X(x) P_{Y|X}(y|x)}{\sum_{z \in \Omega} P_X(z) P_{Y|X}(y|z)}$$

Total Law of Probability

$A = \{X=x\}$ Let $A_1, A_2, \dots$ is a partition of sample space Ω

$$P(A) = \sum_{i=1}^n P(A \cap A_i) = \sum_{i=1}^n P(A|A_i) P(A_i)$$

$$P_X(x) = P(\{X=x\}) = \sum_{i=1}^n P(\{X=x\} | A_i) P(A_i)$$

Let X be a R.V that takes values 1, 2

$$P_{Y|X}(y|1) = \frac{e^{-\lambda_1} \lambda_1^y}{y!} \quad P_X(x) = \begin{cases} p & x=1 \\ 1-p & x=2 \end{cases}$$

$$P_{Y|X}(y|2) = \frac{e^{-\lambda_2} \lambda_2^y}{y!}$$

Find: $P_{X|Y}(1|y)$, $P_{X|Y}(2|y)$

$$P_X(x) = \sum_{i=1}^n P_{X|A_i}(x) P(A_i)$$

Suppose we consider $A_y = \{Y=y\}$, $\forall y \in \mathbb{N}$

$$P_X(x) = \sum_{y \in \mathbb{N}} P_{X|Y=y}(x) P_Y(y)$$

Let $A_1, \dots, A_n$ partition Ω

$$P_{X|A}(x) = \sum P_{X|A \cap A_i}(x) P(A_i|A)$$

$$P_{X|A}(x) = \frac{P(\{X=x\} \cap A \cap \Omega)}{P(A)}$$

$$\Omega = \bigcup_{i=1}^n A_i$$

$$= \frac{\sum_{i=1}^n P(\{X=x\} \cap A \cap A_i)}{P(A)}$$

$$P(A)$$

$$= \sum_{i=1}^n \frac{P(\{X=x\} | A \cap A_i) P(A \cap A_i)}{P(A)}$$

$$\Rightarrow P_{X|A}(x) = \sum_{i=1}^n P(A_i|A) P_{X|A \cap A_i}(x) \quad (*)$$

Conditional expectation

A be an event

$$E[X|A] = \sum_{x \in X_0} x P_{X|A}(x)$$

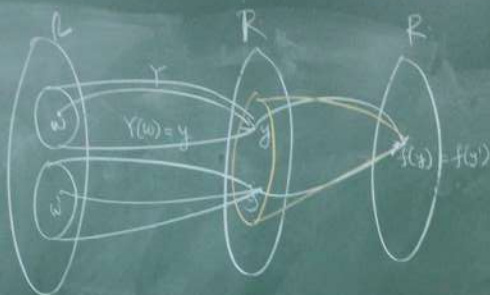
Suppose $A = \{Y=y\}$.

$$\begin{aligned} E[X|Y=y] &= \sum_{x \in X_0} x P_{X|\{Y=y\}}(x) \\ &= \sum_{x \in X_0} x P_{X|Y}(x|y). \end{aligned}$$

Let $f(y) = E[X|Y=y]$

Definition of conditional expectation:

$$E[X|Y] = f(Y)$$



Total Expectation Theorem

Let $A_1, A_2, \dots, A_n$ be a partition of Ω then

$$E[X] = \sum_{i=1}^n P(A_i) E[X|A_i]$$

$$E[X] = \sum_{x \in \mathcal{X}} P_X(x) x = \sum_{x \in \mathcal{X}} \left[\sum_{i=1}^n P_{X|A_i}(x) P(A_i) \right] x$$

$$E[X] = \sum_{i=1}^n P(A_i) \underbrace{\sum_{x \in \mathcal{X}} x P_{X|A_i}(x)}_{E[X|A_i]} = \sum_{i=1}^n P(A_i) E[X|A_i]$$

Suppose $A_Y = \{Y=y\}$ then $\bigcup_{y \in \mathcal{Y}} A_Y = \Omega$. Then

$$E[X] = \sum_{y \in \mathcal{Y}} P(A_Y) E[X|A_Y]$$

$$= \sum_{y \in \mathcal{Y}} P(Y=y) E[X|Y=y]$$

$$= \sum_{y \in \mathcal{Y}} P_Y(y) f(y)$$

$$= E[f(Y)]$$

$$= E[E[X|Y]] \Rightarrow E[X] = E[E[X|Y]]$$

Similarly. $\Rightarrow E[Y] = E[E[Y|X]]$

Example: $P_X(x) = \begin{cases} p & x=1 \\ 1-p & x=2 \end{cases}$

$$P_{Y|X}(y|i) = \frac{e^{-\lambda_i} \lambda_i^y}{y!}$$

$$Y|X=1 \sim \text{Poisson}(\lambda_1)$$

$$Y|X=2 \sim \text{Poisson}(\lambda_2)$$

$$E[Y|X=1] = \lambda_1$$

$$E[Y|X=2] = \lambda_2$$

Find $E[Y] = E[E[Y|X]] = \sum_{i \in \mathcal{X}_0} P_X(i) E[Y|X=i]$

Let $\hat{f}(x) = E[Y|X=x]$

$$= P_X(1) \hat{f}(1) + P_X(2) \hat{f}(2)$$


$$E[Y] = p\lambda_1 + (1-p)\lambda_2$$

$$E[X] = \sum_{i=1}^n P(A_i) E[X|A_i]$$

$$E[X] = E[E[X|Y]]$$

$$E[Y] = E[E[Y|X]]$$

$$E[g(x)] = \sum_{i=1}^n P(A_i) E[g(x)|A_i]$$

$$E[g(x)] = E[E[g(x)|Y]]$$

$$E[h(y)] = E[E[h(y)|X]]$$

Example:

$$\text{Var}(Y) = E(Y^2) - E(Y)^2$$

$$E(Y^2) = E[E(Y^2|X)]$$

for $h(Y) = Y^2$

$$E(Y^2|X=1) = \lambda_1 + \lambda_1^2$$

$$E(Y^2|X=2) = \lambda_2 + \lambda_2^2$$

$$E(Y^2) = p(\lambda_1 + \lambda_1^2) + (1-p)(\lambda_2 + \lambda_2^2)$$

$$\text{Var}(Y) = p(\lambda_1 + \lambda_1^2) + (1-p)(\lambda_2 + \lambda_2^2) - (p\lambda_1 + (1-p)\lambda_2)^2$$

Note: Y is not Poisson. (Correction from class)

If $Z \sim \text{Poisson}(\lambda)$ then

$$E(Z) = \lambda$$

$$\text{Var}(Z) = \lambda$$

$$E(Z^2) = \text{Var}(Z) + (E(Z))^2 = \lambda + \lambda^2$$

(Exercice)

$$P_Y(y) = \sum_{x \in \{1,2\}} P_{X,Y}(x,y)$$

$$= p e^{-\lambda_1} \frac{\lambda_1^y}{y!} + (1-p) e^{-\lambda_2} \frac{\lambda_2^y}{y!}$$

$$\begin{aligned}\text{Var}(X|A) &= E[(X - E[X|A])^2 | A] \\ &= E[X^2 | A] - (E[X|A])^2 \quad [\text{Exercise}] \\ &= \sum_{x \in \mathcal{X}} p_{X|A}(x) (x - E[X|A])^2\end{aligned}$$

$$\begin{aligned}\text{Var}(X|Y=y) &= E[(X - E[X|Y=y])^2 | Y=y] \\ \mu_X &= E[X] \\ \text{Var}(X) &= E[(X - \mu_X)^2] \\ &= E[X^2] - \mu_X^2\end{aligned}$$

Let $v(y) = \text{Var}(X|Y=y)$

$V(Y)$ is a random variable.

Conditional Variance

$$\text{Var}(X|Y) = v(Y) \quad \text{a.r.v.}$$

$$\text{Var}(X) = E[\text{Var}(X|Y)] + \text{Var}[E[X|Y]]$$